

Homivis Avatar Studio

Full Project Report

2 October 2026 · Sessions 1–7 · persona-studio.homivis.com

\$0.65	7	3	Live
Cost per video	Build sessions	AI personas	Platform status

Executive Summary

This project built a fully automated AI influencer video production platform from scratch. Starting with a RunPod GPU pod and ending with a production deployment on Railway at **persona-studio.homivis.com**, we created a system that generates talking-head videos for three British AI personas — Victoria Sterling (wealth/property), Marcus Stone (craftsmanship/property), and Sophia Hart (interior design/luxury) — for approximately **\$0.65 per video**.

The platform covers the entire pipeline: skill-driven script generation via Groq, voiceover synthesis via ElevenLabs, AI portrait and scene generation via FLUX on fal.ai, background removal via rembg, scene compositing, talking-head video generation via fal.ai Flashtalk, grain filtering via ffmpeg, and permanent video storage on fal.ai CDN.

Infrastructure & Architecture

RunPod GPU Pod (Sessions 1–6)

- RTX 4090 pod, 61GB RAM, network volume ID: 0q16vtwovv
- ComfyUI with EchoMimic V3 Flash for animation (commit 6bb7064 pinned)
- FLUX Schnell Q8 GGUF for portrait and scene generation (local, free)
- FastAPI backend on port 8000 (port 8001 for platform)
- Docker image: homivisdocker/avatar-studio:latest
- GitHub repo: souteutobivbi-lab/homivis-avatar-studio

Railway Production Backend (Session 7+)

- FastAPI backend deployed at persona-studio.homivis.com
- Always-on, ~\$5/month, no GPU required
- Custom domain via Cloudflare CNAME
- Single replica (critical for job status consistency)
- Environment variable RAILPACK_DEPLOY_APT_PACKAGES=ffmpeg for ffmpeg
- rembg pre-warms on startup in background thread

APIs in Production

Service	Use	Model/Version	Cost
Groq	Script generation	qwen/qwen3.8-27b	Free
ElevenLabs	Voiceover	eleven_multilingual_v2	~\$0.01/video
fal.ai FLUX	Portraits + scenes	flux/schnell	~\$0.02/image
fal.ai Flashtalk	Talking video	flashtalk 14B	~\$0.60/video
fal.ai storage	Video hosting	CDN	~\$0.001/video
rembg	Background removal	bria model (1GB)	Free (local)

What Worked ✓

FLUX Schnell via fal.ai — Portrait and scene generation. ~\$0.02, ~10 seconds, consistent quality.

rembg bria model — Background removal with clean edges around hair. Best quality tested. 1GB model.

PIL compositing — Portrait onto scene. Instant, free, no dependencies.

fal.ai Flashtalk — Talking-head video from image + audio. ~\$0.60, ~30 seconds. 768x448 output.

ffmpeg grain filter — Skin texture improvement. `noise=all:15:allf=t+u,unsharp=5:5:1.8` — confirmed working.

fal.ai CDN storage — Permanent video URLs that survive Railway restarts.

ElevenLabs TTS — High quality voiceover, voice preview, 10+ voice options.

Groq Qwen — Script generation. Free, fast, good quality. 5 options generated per request.

Persona skill files — Auto-parsed by AI to fill all persona fields. Shapes script tone and vocabulary.

EchoMimic V3 Flash — Animation worked on RunPod with `pose_03`, `motion_sync=True`, `length=375`. Loses identity.

fal.ai sync-lipsync — Video-to-video lip sync worked well. \$2.10 for 3 minutes.

Showcase homepage — Public-facing gallery. Videos publish with one click.

Railway deployment — Full platform on Railway with custom domain `persona-studio.homivis.com`.

What Didn't Work X

Video Generation Models Tested and Rejected

Model	Problem	Verdict
EchoMimic V3 Flash	Regenerates face — loses identity, changes appearance	Rejected for identity
Wav2Lip	Mouth blending artifacts, visible seam around mouth	Rejected
LatentSync	Garbled face output, completely unusable	Rejected
SadTalker	Erratic head movement, crops to face only, loses body	Rejected
Flashtalk (29s audio)	Hung for 30+ minutes on long audio — timed out	Use 15s max
sync-lipsync/v3 image-to-video	\$16.93 per video — discovered late, very expensive	Too expensive
VACE/Fun Control	Changed identity and colours significantly	Rejected
fal.ai Wan2.1 i2v	Did not preserve identity in brief test	Needs more testing

Background Removal Approaches That Failed

Approach	Problem
PIL white pixel threshold	Left white halos around hair — looked bad on composites
rembg bria 1GB on Railway	Downloaded 1GB on every restart — blocked pipeline for minutes
rembg u2net 176MB	Still downloaded on every restart — same blocking issue
nixpacks.toml pre-bake	Railway switched to railpack — completely ignored
Dockerfile pre-bake	Railway ignored Dockerfile when railpack was present
fal.ai birefnet	Added cost and extra failure point to pipeline
apt.txt	Railpack ignored it
Procfile	Railpack ignored it

Not Tested / Partially Tested ■

Feature	Status	Notes
LoRA training for identity	Not tested	Replicate/fal.ai ~\$0.60 one-time — would lock facial identity
fal-ai/instantid	Not tested	Inject reference face into any generated scene
fal-ai/wan/image-to-video	Barely tested	Very cheap \$0.0001 — identity preservation unknown
VEED Fabric 1.0	Not tested	\$0.08/sec — natural gestures but expensive
fal-ai/musetalk	Partially tested	Requires source video not image — different workflow
Cloudflare R2 storage	Not implemented	Free 10GB — better than fal.ai CDN for archiving
Aspect ratio in backend	UI built only	Backend does not pass width/height from ratio selector
EchoMimic with driving video	Partially tested	Better results but no clean 15s driving video available
10-portrait LoRA generation	UI built	Portrait panel built — LoRA training not wired
Direct TikTok/Instagram publish	Not started	API integration would enable auto-publishing
PDF skill file parsing	Basic only	TXT works well — PDF parsing needs more testing

Solutions Found Along the Way

ffmpeg on Railway

Tried nixpacks.toml, apt.txt, Dockerfile, Procfile — all ignored by railpack. Solution: single environment variable RAILPACK_DEPLOY_APT_PACKAGES=ffmpeg.

Script generation model

Claude API too expensive, Groq llama-3.1-70b decommissioned mid-session. Solution: qwen/qwen3.8-27b — free, available, excellent quality.

Video too expensive

sync-lipsync/v3/image-to-video was wired at \$16.93/video without checking cost. Discovered from billing. Solution: switched to Flashtalk at \$0.60/video — 28x cheaper.

Job status lost on Railway restart

Jobs stored in memory reset on every deploy. Solution: store job status in /app/jobs.json — persists across restarts. Also store final video URL on fal.ai CDN.

rembg model download blocking pipeline

Both bria (1GB) and u2net (176MB) downloaded on every Railway restart, blocking the pipeline for minutes. Solution: pre-warm in background thread on startup — downloads once, ready before first user request. Also tried Dockerfile baking (railpack ignored it).

Port exposure on RunPod

Port 8001 not accessible from browser — UI could not reach backend. Solution: migrated entire backend to Railway, eliminating RunPod dependency for everything except local FLUX generation.

EchoMimic losing identity

EchoMimic V3 Flash regenerates the face from scratch rather than animating the input. Tried facemask_ratio=0.0 which helped but did not fully solve it. Solution: switched to Flashtalk which is primarily a lip-sync model.

Composite destroyed by EchoMimic

Sending a pre-composited portrait (portrait + scene) to EchoMimic resulted in EchoMimic stripping the background and regenerating both face and background. Root cause: EchoMimic crops to face region regardless of input. Solution: composite AFTER video generation, not before.

Cost Analysis

Current Cost Per Video

Step	Service	Cost
Portrait generation	fal.ai FLUX Schnell	~\$0.02
Scene generation	fal.ai FLUX Schnell	~\$0.02
Script generation	Groq qwen/qwen3.8-27b	Free
Voiceover	ElevenLabs multilingual v2	~\$0.01
Background removal	rembg bria (local)	Free
Talking video	fal.ai Flashtalk 14B	~\$0.60
Video storage	fal.ai CDN	~\$0.001
TOTAL		~\$0.65

Scale Economics

Volume	Cost/month (videos only)	Infrastructure	Total/month
30 videos/month	\$19.50	\$5 Railway	~\$25
90 videos/month (3 personas)	\$58.50	\$5 Railway	~\$64
300 videos/month	\$195	\$5 Railway	~\$200

Recommendations: Realistic Human Portraits

The most important unsolved problem: how to generate a portrait that looks like a real, specific human and maintains that identity consistently across all videos and scenes.

Option 1 — LoRA Training (Best for Identity Lock)

Train a custom LoRA on 10–20 images of the persona using `fal-ai/flux-lora-fast-training`.

- Generate 10 portraits with FLUX in different outfits and angles
- Train a LoRA: `fal-ai/flux-lora-fast-training` (~\$0.60 one-time per persona)
- Use the LoRA trigger word in all future FLUX generations — same face every time
- Result: consistent facial identity across all images, scenes and videos
- Cost: ~\$0.60 once per persona, then free on all subsequent portrait generations

Option 2 — Higgsfield AI (Best Current Quality)

The Higgsfield-generated Victoria Sterling portrait was the most realistic image produced in the entire project. Natural skin, real pores, authentic expression, consistent identity. Higgsfield is connected via MCP in this Claude conversation and can generate portraits directly.

- Use Higgsfield to generate high-quality portrait in multiple outfits
- Lock the best portrait as the permanent persona reference image
- Do not regenerate — use the same image for all videos
- Higgsfield also offers LoRA-style consistent character generation

Option 3 — Real Photography (Maximum Realism)

For truly indistinguishable results, photograph a real person once and use that image permanently.

- White background, professional studio lighting
- Upper body, front-facing, natural expression
- High resolution (minimum 2MP)
- Real photos preserve identity perfectly in Flashtalk and other models
- One shoot covers hundreds of videos

Option 4 — `fal-ai/instantid` (Identity Injection)

InstantID takes a reference face photo and applies it to any generated image or scene.

- Have one master portrait photo (real or Higgsfield)
- Use `fal-ai/instantid` to generate new images with same face in any outfit or scene
- All outputs have consistent facial identity regardless of prompt
- Not yet tested — high priority for next session

Next Steps

Immediate (This Week)

- Fix Flashtalk video aspect ratio — output is 768x448, reframe to show full upper body properly
- Train LoRA for Victoria Sterling using Higgsfield portraits
- Test fal-ai/instantid for identity-consistent scene images
- Test fal-ai/wan/image-to-video as a cheaper Flashtalk alternative (\$0.0001 vs \$0.60)
- Fix aspect ratio parameter passing from UI to backend
- Confirm pipeline runs end-to-end cleanly without any blocking issues

Short Term (Next 2 Weeks)

- Wire LoRA trigger word into portrait generation for all three personas
- Generate 10 portraits each for Marcus Stone and Sophia Hart
- Build content calendar — batch 30 scripts per persona
- Add video thumbnail generation for showcase
- Move video storage to Cloudflare R2 (free 10GB vs fal.ai CDN paid)

Medium Term (Next Month)

- Add TikTok and Instagram API for direct publishing from the platform
- Add video analytics to showcase homepage
- Build Marcus Stone and Sophia Hart complete persona profiles with LoRA
- Add music/audio bed option for videos
- Consider adding captions auto-generation (fal-ai/whisper)

Platform vision: 30 videos per month per persona, published automatically to TikTok and Instagram, at ~\$0.65/video = ~\$20/month in API costs. With three personas that is ~\$60/month for 90 pieces of content. The platform is working — the priority is now reliable identity preservation to make output indistinguishable from real influencer content.